Daily Experimental Findings: Frame-Invariant Forecasting with GRPO

InvariantForecasting Project

April 30, 2026

Abstract

This note summarizes the experimental findings from the April 30, 2026 RunPod session for the InvariantForecasting project. The core question was whether GRPO-style post-training on ForecastBench paraphrase groups can reduce forecast instability across semantically equivalent question variants while preserving ordinary forecasting accuracy. The strongest clean result observed today was a Mantic-style GRPO run with training invariance weight $\lambda_{\text{train}} = 1$, selected by validation checkpoint performance and evaluated once on the held-out test split. It reduced test paraphrase dispersion from 0.08434 to 0.08083 and slightly reduced FRIE-1 from 0.32031 to 0.31938, while increasing Brier score from 0.23597 to 0.23854. A stronger $\lambda_{\text{train}} = 2$ run looked better on validation at checkpoint 10 but did not transfer to test, where paraphrase dispersion and FRIE-1 worsened. The main conclusion is that the implementation is now much safer and the research signal is plausible but still fragile. The next scientific step is replication across seeds and a prespecified checkpoint-selection rule.

1 Purpose of the Session

The session had three practical goals:

1. Verify that the GRPO implementation can train and evaluate without collapsing into degenerate completions such as repeated punctuation.
2. Compare the base Qwen2.5-7B-Instruct forecaster against invariance-regularized GRPO variants.
3. Determine whether the observed paraphrase-invariance improvements are strong enough to justify a workshop-paper direction.

The most important methodological constraint was to distinguish a theoretical failure from an implementation failure. In other words, if the model fails to improve, that failure should be attributable to the training objective or data regime, not to a broken reward parser, unstable RL configuration, or invalid checkpoint.

2 Dataset and Evaluation Setup

The data are ForecastBench current-event questions filtered to questions resolved after Qwen's relevant knowledge cutoff, with five paraphrase variants per underlying question group.

Split	Groups	Variants	Notes
Train	478	2,390	Used for GRPO post-training
Validation	60	300	Used for checkpoint selection
Test	60	300	Used for final held-out comparison

Each group contains five paraphrased variants of the same forecasting question. The evaluation reports both per-variant forecast accuracy and within-group paraphrase instability.

3 Metrics

The key metrics used today were:

- **Brier score**: lower is better. This measures forecast accuracy against the realized binary outcome.
- **ParaSD**: lower is better. This measures dispersion of predicted probabilities across paraphrases in the same question group.
- **FRIE-1**: lower is better. This is the combined metric with invariance penalty weight 1, conceptually
$$\mathrm{FRIE}_1 = \mathrm{Brier} + \mathrm{ParaSD}.$$
- **Coverage**: should be 1.0. This confirms all variants/groups were parseable and included in the metric computation.

Because the test split has only 60 groups, small absolute differences should be treated as directional evidence until replicated across seeds.

4 Implementation Findings

4.1 Original GRPO Collapse

The original full lambda0 run was not merely a weak model. It was invalid. The model learned to output repeated exclamation marks, yielding completions like:

```
!!!
```

4.2 Safety Improvements

The safer GRPO path introduced or relied on the following checks:

- parse-rate monitoring during training;
- punctuation-loop detection;
- non-finite gradient checks;
- conservative learning rate, KL regularization, low sampling temperature, and gradient clipping;
- preflight validation of split sizes, prompt rendering, token lengths, and GRPO group layout.

The later Mantic-style runs completed with full parseability:

- `variant_coverage = 1.0`
- `n_parseable_groups = 60`
- `n_variant_predictions = 300`

This substantially increases confidence that the remaining results reflect the objective and data regime rather than an obvious implementation crash.

5 Main Results

5.1 Held-Out Test Comparison

The most important comparison is the held-out test split. The base model was plain Qwen2.5-7B-Instruct without GRPO post-training. Lower is better for Brier, ParaSD, and FRIE-1.

Run	Brier ↓	ParaSD ↓	FRIE-1 ↓	Coverage
Base Qwen2.5-7B-Instruct	0.235966	0.084341	0.320307	1.0
Mantic GRPO $\lambda_{\text{train}} = 1$, ckpt20	0.238545	0.080833	0.319378	1.0
Mantic GRPO $\lambda_{\text{train}} = 2$, ckpt10	0.235530	0.085824	0.321354	1.0

5.2 Test Delta Relative to Base

Run	ΔBrier	ΔParaSD	ΔFRIE-1
Mantic GRPO $\lambda_{\text{train}} = 1$, ckpt20	+0.002578	-0.003507	-0.000929
Mantic GRPO $\lambda_{\text{train}} = 2$, ckpt10	-0.000437	+0.001483	+0.001047

The lambda1 run produced the most relevant positive result: a meaningful reduction in paraphrase dispersion and a small improvement in FRIE-1, at the cost of worse Brier score. The lambda2 run slightly improved Brier but worsened both paraphrase dispersion and FRIE-1 on test.

6 Validation Checkpoint Selection

Checkpoint selection was performed on validation, not on test. This matters because RL checkpoints are not monotonic: later checkpoints are more trained, but not necessarily better. The validation split was used to select a checkpoint before test evaluation.

6.1 Mantic GRPO, $\lambda_{\text{train}} = 1$

Checkpoint	Brier ↓	ParaSD ↓	FRIE-1 ↓
Base validation	0.240697	0.084712	0.325409
ckpt5	0.240153	0.085576	0.325729
ckpt10	0.239263	0.080787	0.320050
ckpt15	0.236637	0.091164	0.327800
ckpt20	0.234193	0.084778	0.318971
ckpt25	0.235847	0.092573	0.328419
ckpt30	0.235615	0.085031	0.320646

The validation-selected checkpoint by FRIE-1 was ckpt20. On test, ckpt20 reduced ParaSD and slightly improved FRIE-1, but worsened Brier.

6.2 Mantic GRPO, $\lambda_{\text{train}} = 2$

Checkpoint	Brier ↓	ParaSD ↓	FRIE-1 ↓
Base validation	0.240697	0.084712	0.325409
ckpt5	0.237883	0.086303	0.324187
ckpt10	0.236280	0.080537	0.316817
ckpt15	0.235993	0.088916	0.324909
ckpt20	0.239687	0.091482	0.331169
ckpt25	0.238705	0.094720	0.333425
ckpt30	0.241588	0.080897	0.322485

The validation-selected checkpoint by FRIE-1 was ckpt10. However, ckpt10 did not transfer positively to test: test Brier improved slightly, but ParaSD and FRIE-1 worsened. This is evidence that the stronger invariance weight may overfit the validation paraphrase structure or perturb calibration in a way that does not generalize.

7 Interpretation

7.1 What Was Discovered

The session produced four concrete findings.

1. **The original collapse was an implementation/training failure.** The repeated-exclamation behavior was not a meaningful learned forecasting policy. It was a degenerate RL failure mode caused by parse failures and unstable reward dynamics.

2. **The safer GRPO implementation can run cleanly.** Later Mantic-style runs reached full evaluation coverage with parseable outputs and no observed punctuation-loop collapse.

3. **A lambda1 invariance objective can reduce test paraphrase sensitivity.** The best clean test result reduced ParaSD from 0.08434 to 0.08083. This is the strongest positive empirical signal from the day.

4. **More invariance pressure is not automatically better.** Lambda2 looked excellent on validation but failed to improve test FRIE-1. This is a useful negative result and should shape the next experimental design.

7.2 Scientific Strength of the Result

The result is promising but not yet conclusive. The lambda1 test improvement in ParaSD is aligned with the research hypothesis, and the fact that the evaluation remained fully parseable makes it usable. However:

- the test split contains only 60 groups;
- seed variance is visible in earlier lambda1 replications;
- Brier sometimes worsens when ParaSD improves;
- validation improvements do not always transfer to test;
- no confidence intervals or bootstrap tests have been computed yet.

The appropriate claim at this point is therefore:

> In a controlled ForecastBench paraphrase setting, invariance-regularized GRPO can reduce paraphrase sensitivity in some runs while preserving parseability, but the effect is modest and seed/checkpoint sensitive.

This is a plausible workshop-paper seed, not yet a completed ICML-level result.

8 Recommended Next Steps

8.1 Immediate Experimental Steps

1. Preserve the current RunPod artifacts: `results/`, `logs/`, `configs/`, and `outputs/best/`.
2. Run bootstrap confidence intervals over groups for Brier, ParaSD, and FRIE-1.
3. Replicate the Mantic lambda1 and lambda2 setups over multiple seeds.
4. Select checkpoints by validation FRIE-1 only, then evaluate exactly once on test.
5. Inspect qualitative examples where ParaSD improved and where it worsened.

8.2 Paper-Framing Steps

If replications preserve the direction of the lambda1 result, a credible workshop paper could focus on:

- paraphrase instability as a measurable forecasting reliability problem;
- ForecastBench paraphrase groups as a compact evaluation bed;
- GRPO with an invariance penalty as a lightweight intervention;
- the observed tradeoff between calibration/accuracy and paraphrase invariance;
- negative results showing that stronger invariance penalties can fail to generalize.

9 Bottom Line

Today did discover something real. The most important result is not that the method is already solved, but that the project now has:

- a known invalid failure mode and safeguards against it;
- a clean evaluation pipeline with full parseability;
- a positive lambda1 test signal on paraphrase dispersion;
- a negative lambda2 generalization result that constrains the theory.

The project has enough signal to justify continued experimentation and possibly the beginning of a workshop-paper outline, provided the next round focuses on replication, uncertainty estimates, and careful checkpoint-selection protocol.